SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I SOMAVARSHINI.M (192...) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Somavarshini.M

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	6
Experimental Design	30	7
Use of Tools	20	5
Analysis of Results	15	4
Documentation	10	2

Total Marks 24 **/30**

Signature of the Faculty

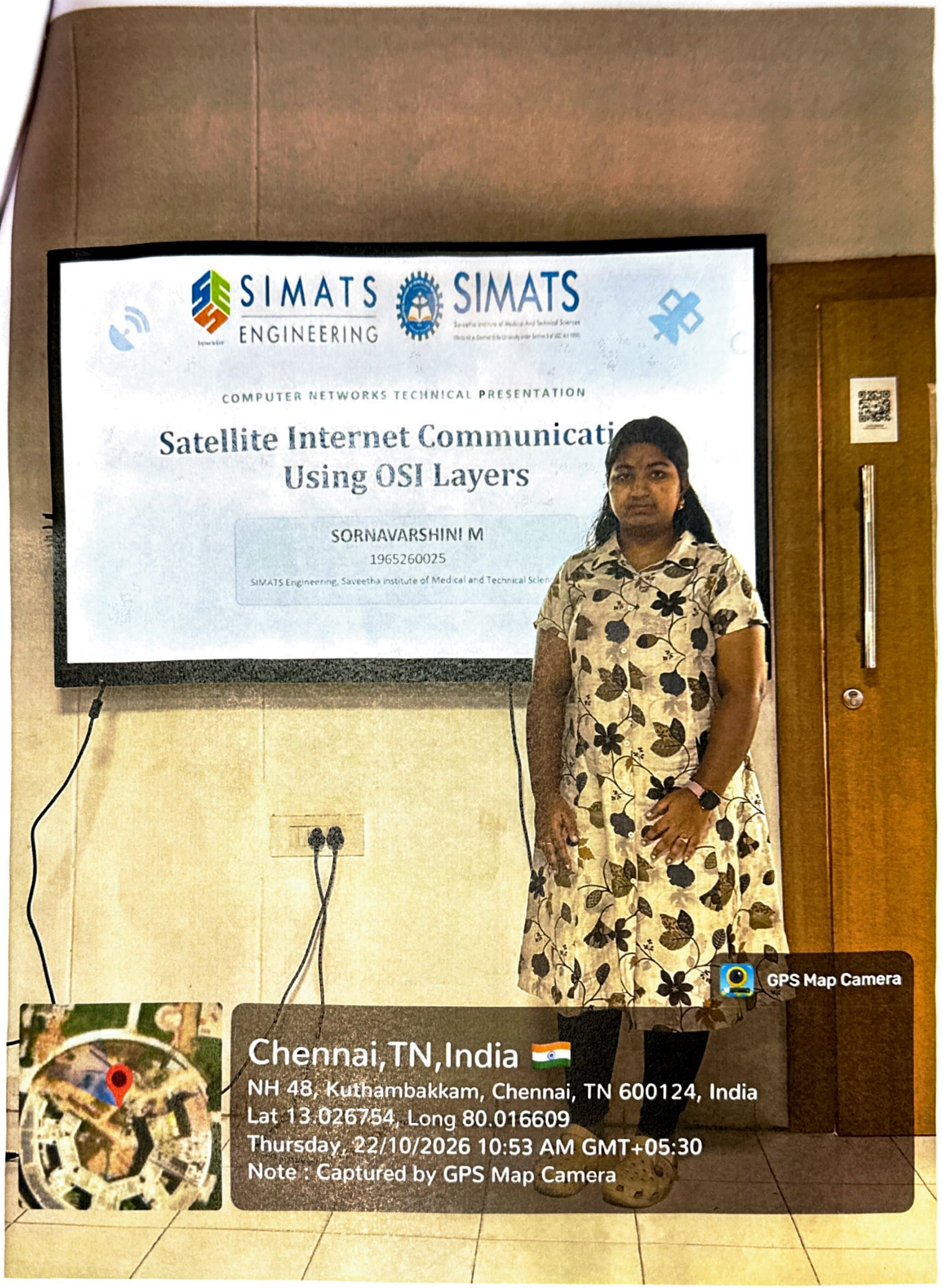
TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretatio n		
Documentati on	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentati on with necessary details	Basic documentati on with missing details	Poor or incomplete documentati on


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COMPUTER NETWORKS TECHNICAL PRESENTATION
Satellite Internet Communicati
Using OSI Layers
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Chennai,TN,India
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SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

COMPUTER NETWORKS TECHNICAL PRESENTATION

Satellite Internet Communication Using OSI Layers

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SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences

Overview

- What: internet access in which data is relayed between a user and the internet through an orbiting satellite.
- Why it matters: it reaches places where fibre or cable is impractical or too costly to build.
- OSI view: the seven layers separate the work, from application data down to the radio signal.



User terminal

Device, router and dish at the user's site



Satellite

Relays radio signals in orbit



Ground gateway

Links the satellite to terrestrial networks



Internet server

Hosts the requested content or service

Introduction

Satellite internet: a service that carries IP traffic between a user and the internet over radio links relayed by satellites.

How data travels

1. Device sends IP packets to the user's terminal
2. Dish transmits them by radio to the satellite
3. Satellite relays them to a ground gateway
4. Gateway forwards them into the internet

Aspect	Satellite	Terrestrial
Medium	Radio through space	Fibre, copper, cable
Coverage	Wide, even remote	Where lines are laid
Delay	Higher (GEO most)	Generally lowest

Applications


Remote areas


Ships


Aircraft


Disaster recovery


Rural connectivity

Problem Statement



Coverage gaps

Terrestrial networks are costly or impractical in remote and rural regions.



Reliable connectivity

Remote users still need stable, always-available internet access.



Satellite link issues

Latency, signal interference, weather effects and packet loss reduce performance.

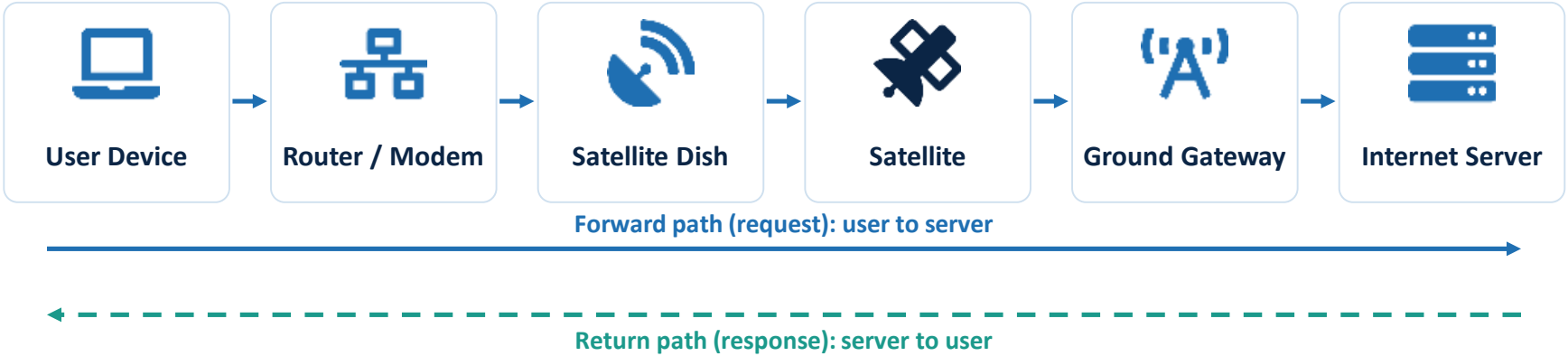


Layer coordination

A fault can occur at any layer, so we must know how the layers cooperate.

Key question: how do the OSI layers coordinate to deliver data over a satellite link?

System Architecture and Methodology



Simplified bent-pipe model; some LEO systems also use inter-satellite links.

Methodology

1. **Identify network components**
2. **Map functions to OSI layers**
3. **Trace data transmission**
4. **Analyze potential failures**

OSI Model Overview

Layer	Role in satellite internet	Data unit
7 Application	Web, DNS and streaming services (HTTP/HTTPS)	Data
6 Presentation	Data formats, encryption (e.g. TLS), compression	Data
5 Session	Opens, manages and closes application dialogues	Data
4 Transport	End-to-end delivery: TCP reliability, UDP speed	Segment / Datagram
3 Network	IP addressing and routing via the gateway	Packet
2 Data Link	Framing, link access and error detection per link	Frame
1 Physical	RF signals, modulation, antennas, satellite link	Bits / Signal

Application, Presentation, Session Layers

Layer 7: Application

- HTTP/HTTPS for web access
- DNS maps names to IP addresses
- User-facing services: browsing, email, video calls

Supports: page loads and streaming requests

Layer 6: Presentation

- Data formats, e.g. JPEG, JSON, UTF-8
- Encryption, e.g. TLS
- Compression, e.g. video codecs

Supports: secure, efficient content

Layer 5: Session

- Starts, maintains and ends dialogues
- Often handled inside application protocols
- Examples: login sessions, call signalling

Supports: calls and continuous streams

Note: TCP/IP merges layers 5 to 7 into one Application layer. Many apps (e.g. HTTPS = HTTP + TLS) do not use three separate protocols.

Transport and Network Layers

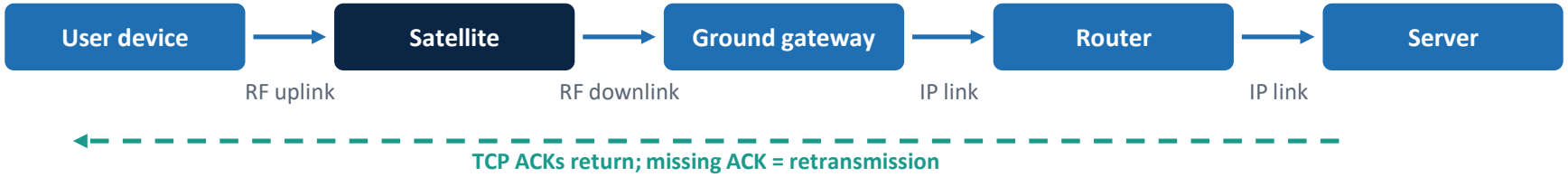
Layer 4: Transport

- TCP: ordered, reliable delivery using sequence numbers, ACKs and retransmission
- UDP: low overhead, no retransmission (e.g. DNS, VoIP, QUIC)
- High delay slows TCP; loss triggers retransmits

Layer 3: Network

- IP addresses identify source and destination
- Routers and the gateway forward packets hop by hop
- Delay and loss add up end to end

Packet flow



LEO (~550 km): typically tens of ms

GEO (35,786 km): ~500 ms+ round trip

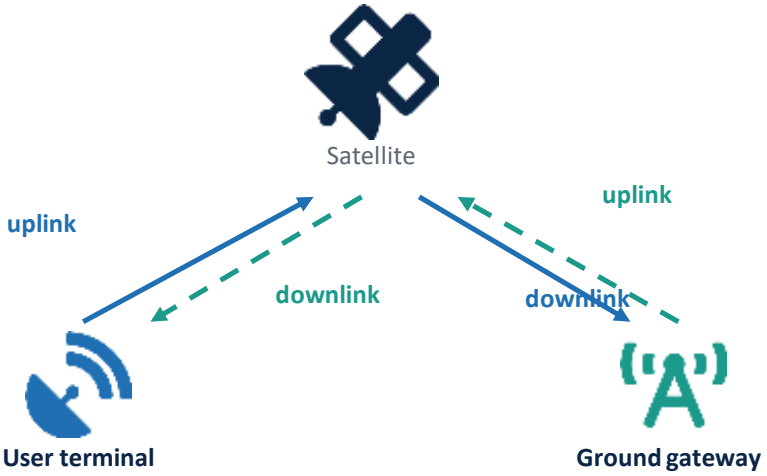
Data Link and Physical Layers

Layer 2: Data Link

- Framing of packets for each link
- MAC addressing where the link uses it (e.g. Ethernet, Wi-Fi)
- Error detection (e.g. CRC) and link-layer coding

Layer 1: Physical

- Radio-frequency carriers and modulation
- Antennas, amplifiers and the satellite link



Solid blue: forward (user uplink, gateway downlink)
Dashed green: return (gateway uplink, user downlink)

Note: the satellite hop uses specialized link protocols (e.g. DVB-S2 framing or provider-specific schemes), not plain Ethernet end to end.

Advantages and Limitations

Advantages

- Wide geographical coverage
- Connectivity in remote locations
- Rapid deployment without terrestrial infrastructure
- Useful for emergency and disaster communications

Limitations

- Higher latency in GEO systems
- Weather-related attenuation in some bands
- Equipment and service costs
- Limited capacity or congestion in some systems
- Potential packet loss and performance variation

Conclusion and References

- The OSI model separates satellite internet into layers, each with a clear task.
- Physical, Data Link, Network, Transport and upper layers work together end to end.

OSI-based troubleshooting

L1 Signal	dish alignment, rain fade
L2 Link	sync loss, frame errors
L3 Routing	IP, gateway, route faults
L4 Delivery	loss, retransmissions
L7 Application	DNS or HTTP errors

References

1. ISO/IEC 7498-1:1994, Open Systems Interconnection: Basic Reference Model. https://www.iso.org/standard/20269.html

2. Cisco Networking Academy, networking and OSI model courses. https://www.netacad.com

3. NASA, Space Communications and Navigation (SCaN). https://www.nasa.gov/scan

4. ITU-R, Radiocommunication Sector: satellite orbits and spectrum. https://www.itu.int/en/ITU-R/information/Pages/default.aspx

5. Starlink Specifications (latency figures). https://www.starlink.com/legal/documents/DOC-1402-50343-58

6. IETF RFC 9293, Transmission Control Protocol (TCP). https://www.rfc-editor.org/info/rfc9293

7. IETF RFC 2488, Enhancing TCP Over Satellite Channels. https://www.rfc-editor.org/info/rfc2488

THANK YOU!

Any Questions?

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